MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 6014

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, inert gas

See figure 1

Mounting position: Any

Weight: 3 ounces (85 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	Ib	ib	tk	TA	F	i(surge) 0.1 sec
Unit:	Vac	v	v	Adc	a	sec	°C	Hz	a
Maximum:	2.63	1,250	1,000	1.0	8.0	---	+75	1,000	77
Minimum:	2.37	---	---	---	---	20	-60	---	---
TEST CONDITIONS:	2.5	1,250	1,000	---	---	---	---	60	---

GENERAL:

Qualifications - Required

Stabilization (MIL-STD-1311)
(t = 16 hours; Epp = 130 Vac (max);
Ib = 1.2 Adc ±10%; Rg = 10,000 ohms,
grid connected to anode)

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3201	Critical grid voltage for conduction	-	Ebb = 1,000 Vdc; Rg = 1,000 ohms; Rp = 750 to 5,000 ohms	0.65	II	Ecc	-2.5	-6.5	Vdc
3201	Critical anode voltage for conduction	-	Ec = 4 Vdc; Rg = 1,000 ohms; Rp = 750 to 5,000 ohms	0.65	II	Ebb	---	75.0	Vdc
3204	Voltage drop:	1	Ib = 1.0 Adc; Epp = 130 Vac (max); Rg = 10,000 ohms; grid connected to anode	0.65	II	Etd	---	12.0	Vdc
3206	Operation	-	Ib = 1.0 Adc; Rg = 10,000 ohms	0.65	II	Ecc	-2.5	-6.5	Vdc
3216	Grid current	2	Rg = 1.0 Meg and 10,000 ohms	0.65	II	ic	---	5.0	µa
1231	Emission	3	ib = 8a	0.65	II	etd	---	45.0	v
	Quality conformance inspection, part 2								
3241	Heater current			---	---	If	5.5	7.1	Aac
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1042	Shock, specified pulse	6	Test condition A	---	---	---	---	---	---
----	Filament cycling (no load)	5	Eb = Ec = 0; t = 1,000 cycles	---	---	---	---	---	---
	Quality conformance inspection, part 3								
----	Life test	-	Group D; operation; ib = 8a; Ib = 1.0 Adc; resistance load; t = 500 hrs	---	---	---	---	---	---
---	Life-test end points:								
3206	Operation	-	Ib = 1.0 Adc; Rg = 10,000 ohms	---	---	Ecc	-2.5	-6.5	Vdc
3216	Grid current	2	Rg = 1.0 Meg and 10,000 ohms	---	---	ic	---	5.0	µa
1231	Emission	3	ib = 8a	---	---	etd	---	55.0	v
3216	Grid current	4	epx = epy = 750 v; F = 500 Hz; Ib = 1.0 Adc	---	---	ic	---	25.0	µa

NOTES:

1. Wattmeter method: With an alternating current source and resistive load, measure the power loss in the tube with a wattmeter so that the current coil is in series with, and the voltage coil across, the tube. The tube voltage drop (Etd) is the total power loss minus the coil loss under load, divided by the average, or dc, current (Ib).

2. Operate tube at least 5 minutes before the test. Operation may be at optional Epp, but Ib shall be 1.0 Adc minimum. To test, switch to the test voltage and Ib = 1.0 Adc in less than 2 seconds.

3. Connect grid to anode through 10,000-ohm resistor. Adjust Epp and Rp to give the required current. This test to be the first test performed at the conclusion of the holding period.

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(F) 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

5. A cycle shall consist of a 3-minute period with filament voltage applied followed by a 3-minute period with filament voltage removed. After cycling test, the tube shall be within the limits specified for operation and grid-current tests.

(F) 6. The tube shall meet the limits of all tests specified in quality conformance inspection, part 1.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

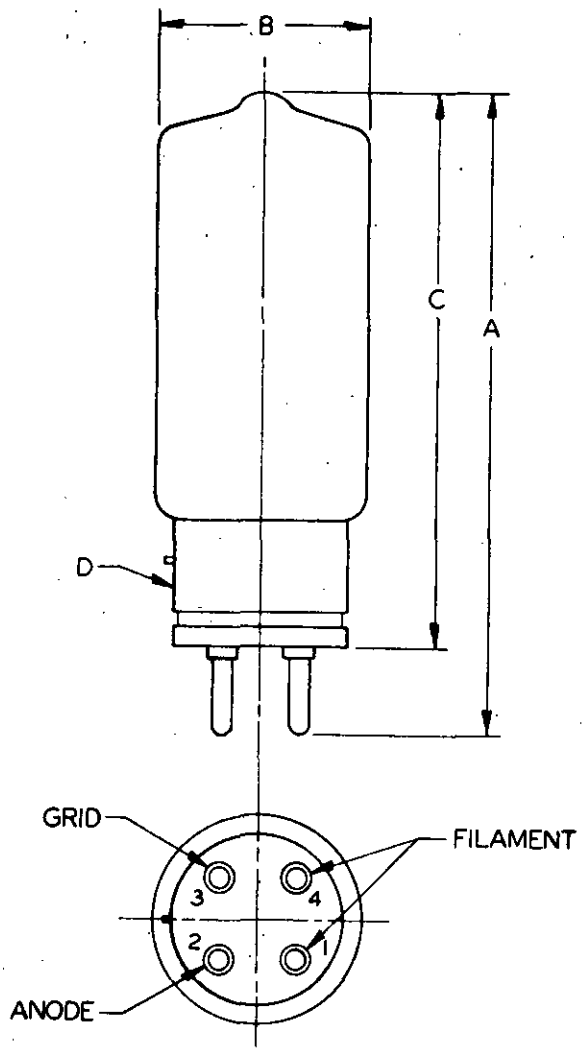
User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3077)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.31 (109.47)
B		1.56 (39.62)
(F) Quality conformance inspection, part 3 (see note)		
C		3.78 (96.01)
D	Base: A4-10 (EIA)	

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6014.

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